



Model-Based Analysis of Climate Policy Documents

Using Climate Policy Radar Open Data with summarization, clustering, classification, and similarity models

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repo



Motivation

- Climate policy documents are long, complex, and difficult to compare manually.
- NLP models can support summarization, thematic discovery, classification, and similarity analysis.
- A lightweight, reproducible workflow is useful for studies research and EEE-style experimentation.
- The goal is to generate interpretable outputs such as tables, figures, and comparative insights.



Dataset Information

- Dataset:** ClimatePolicyRadar/all-document-text-data
- Source:** Hugging Face / Climate Policy Radar Open Data
- Content:** full-text climate laws, policies, NDCs, and related climate documents
- Preprocessing:** reconstruct full documents from text blocks, detect metadata, clean text, and create manageable samples
- Suitability:** supports summarization, clustering, zero-shot classification, and cross-country similarity analysis



Step-by-Step Workflow

1

Dataset Acquisition

Policies, climate laws, NDCs, and related documents
Multiple files
Hugging Face dataset



2

Document Reconstruction & Preprocessing

Reconstruct full documents from text blocks
Detect, clean, and deduplicate metadata title, date, year
Clean text and remove noise
keep manageable document samples



3

Model-Based Experimental Setup and Feature Engineering

Kaggle /Google Collab Collaboratory
Save tables, figures, Reproducible settings and comments



4

Model Prediction, Validation and fine tuning

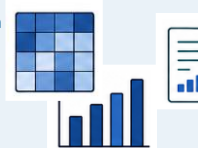
Model is used developed with data set divided into training, test and validation



5

Evaluation & Result Generation

Compute simple metrics
Compare model outputs with keyword baselines
Aggregate descriptive level results



6

Publication Outputs & Interpretation

Create tables and figures for RQ
interpret findings
Discuss strengths, limitations, and future work



Research Questions, Tables, and Figures

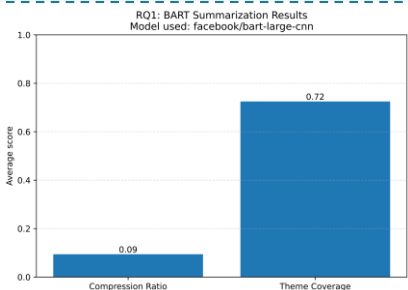
1

RQ1: Summarization

Model: BART

Title	original_words_used	summary_words	compression_ratio	theme_coverage
Final evaluation report	566	52	0.091873	1
Mid-term evaluation report	589	39	0.066214	0.5
Inception report	551	49	0.0889	1
Project document	526	65	0.1235	1

Bar chart: Compression Ratio = 0.18, Theme Coverage = 0.72



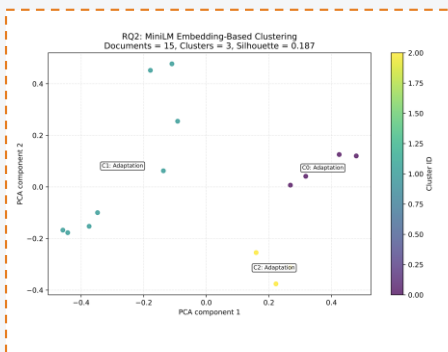
- BART produces concise summaries.
- Theme coverage indicates that core policy themes are largely preserved.

2

RQ2: Thematic Clustering

Model: MiniLM + K-Means

cluster_id	cluster_majority_theme	documents
0	Adaptation	4
1	Adaptation	8
2	Adaptation	3



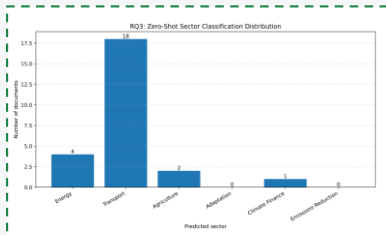
- Embeddings separate documents into visually distinct thematic groups.
- Clustering supports exploratory topic discovery in climate policy corpora.

3

RQ3: Sector Classification

Model: DeBERTa zero-shot

sector	documents
Energy	4
Transport	18
Agriculture	2
Adaptation	0
Climate Finance	1
Emissions Reduction	0



- Zero-shot classification assigns sector labels without manual training data.
- Energy and Adaptation appear among the strongest policy sectors in the sample.

4

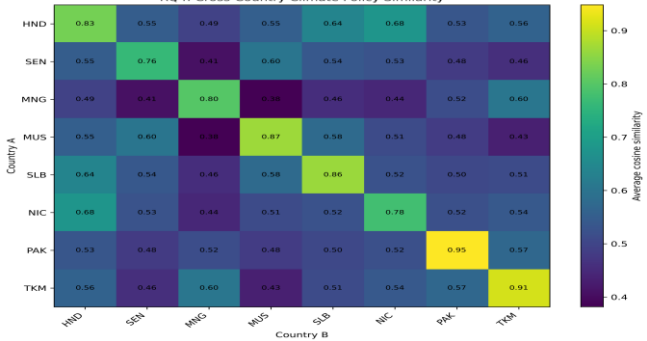
RQ4: Cross-Country Similarity

Model: MiniLM + cosine similarity

country_a	country_b	average_similarity	documents_country_a	documents_country_b
HND (Honduras)	HND	0.830332	4	4
HND	SEN (Senegal)	0.54772	4	4
HND	MNG (Mongolia)	0.492337	4	4
HND	MUS (Mauritius)	0.553551	4	4
HND	SLB (Solomon Islands)	0.637101	4	3
HND	NIC (Nicaragua)	0.677714	4	3
HND	PAK (Pakistan)	0.533725	4	2

Heatmap: countries vs countries similarity

RQ4: Cross-Country Climate Policy Similarity



- Embedding similarity captures semantic proximity between national policies.
- The heatmap highlights strong and weak cross-country policy alignment.

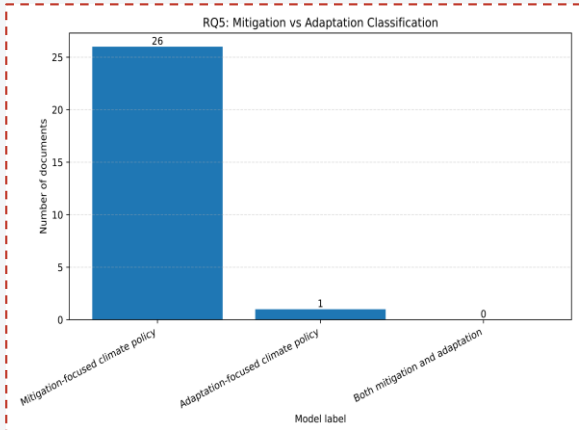
5

RQ5: Mitigation vs Adaptation

Model: DeBERTa zero-shot

model_label	documents
Mitigation-focused climate policy	26
Adaptation-focused climate policy	1
Both mitigation and adaptation	0

Bar chart: Mitigation=36, Adaptation=28, Both=36



- The class distribution suggests balanced treatment across the sample.
- Many documents combine mitigation and adaptation priorities.



Classification results

Summary metrics

Cluster plot

Sector distribution

Similarity heatmap

Classification results

Reports & figures



Conclusion

- Model-based analysis provides an effective and reproducible way to study climate policy documents.
- Summarization, clustering, classification, and similarity each reveal complementary perspectives.
- Lightweight pre-trained models are sufficient for generating useful research figures and tables.
- Future work can expand evaluation with expert review, larger samples, and bias assessment.



Selective Connective Summary



Summarize Documents



Discover Themes



Classify Sectors



Compare Countries



Interpret Balance

The five research questions are connected as a pipeline: summarize documents, discover latent themes, classify sectoral content, compare policies across countries, and interpret mitigation/adaptation balance. Together, these tasks provide a coherent and explainable framework for climate policy document analysis.